SC7 Bayes Methods MT26

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Lecture 11 : MCMC updates using transformations

(preamble to model averaging with Reversible-Jump MCMC)

MCMC using transformations in fixed dimension

Target $\pi(d\theta) = \pi(\theta)d\theta$ with $\theta \in \Omega$.

Take density $g(u)$, $u \in \mathcal{U}$ and function $\psi_1(\theta, u)$ invertible for u at fixed θ. Given θ, set $\theta' = \psi_1(\theta, u)$ with $u \sim g(u)$. Then

$$q(d\theta'|\theta) = g(u)du \tag{1}$$

gives the change of variables under an integral so the density is

$$q(\theta'|\theta) = g(u)\left|\frac{\partial\theta'}{\partial u}\right|^{-1}$$

Example: for $a > 0$ set $\theta' = \theta + a(2u - 1)$ with $u \sim U(0, 1)$,

$$g(u) = \mathbb{I}_{0<u<1}, \quad \psi_1(\theta, u) = \theta + a(2u-1), \quad q(\theta'|\theta) = \frac{\mathbb{I}_{\theta-a<\theta'<\theta+a}}{2a}.$$

If the value $u' \in \mathcal{U}$ reverses the move, so $\psi_1(\theta', u') = \theta$, then

$$\psi_1(\psi_1(\theta, u), u') = \theta. \tag{2}$$

Example: In the example above, check (2) for $u' = 1 - u$.

Let $u' = \psi_2(\theta, u)$ be the function* solving (2). We map pairs

$$(\theta', u') = (\psi_1(\theta, u), \psi_2(\theta, u)).$$

If $\psi = (\psi_1, \psi_2)$ then ψ is an *involution*,

$$(\theta, u) = \psi(\psi(\theta, u)).$$

Example: if $\psi(\theta, u) = (\theta + a(2u - 1), 1 - u)$ as before then

$$\begin{aligned} \psi(\psi(\theta, u)) &= \psi(\theta + a(2u - 1), (1 - u)) \\ &= (\theta + a(2u - 1) + a(2(1 - u) - 1), 1 - (1 - u)) \\ &= (\theta, u) \end{aligned}$$

Up till now we chose $q(\theta'|\theta)$ and found a density $g(u)$ and a function $\theta' = \psi_1(\theta, u)$ to simulate it. Let's just write down g and ψ_1 and let q be whatever it is. This gives more freedom.

*We must choose ψ_1 so a solution exists and ψ_2 is continuously differentiable.

Theorem: let $\psi : \Omega \times \mathcal{U} \to \Omega \times \mathcal{U}$ be an invertible, differentiable involution with $\theta, \theta' \in \Omega$ and $u, u' \in \mathcal{U}$.

The MCMC update with proposal $u \sim g(u)$, $(\theta', u') = \psi(\theta, u)$, acceptance ratio

$$r(\theta', u'|\theta, u) = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u), \tag{3}$$

with J_ψ the Jacobian for the transformation $(\theta', u') = \psi(\theta, u)$,

$$J_\psi(\theta, u) = \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right|,$$

and acceptance probability

$$\alpha(\theta'|\theta) = \min\left\{1,\ r(\theta', u'|\theta, u)\right\} \tag{4}$$

satisfies DB in Eqn 7 with q given in Eqn 1.

Proof: Let

$$K(\theta, d\theta') \;=\; \alpha(\theta'|\theta)q(d\theta'|\theta) \;+\; c(\theta)\delta_\theta(d\theta') \tag{5}$$

with c the probability to reject,

$$c(\theta) = \left[1 - \int_\Omega q(d\theta'|\theta)\alpha(\theta'|\theta)\right].$$

K is the transition kernel K for the MCMC. DB is

$$\int_B \pi(d\theta') \int_A K(\theta', d\theta) = \int_A \pi(d\theta) \int_B K(\theta, d\theta') \tag{6}$$

for all measureable A, B. If this holds the process is stationary,

$$\pi(B) = \int_\Omega \pi(d\theta)K(\theta, B).$$

Sub (5) into (6). The $c(\theta)$-term cancels, so DB is

$$\pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta') = \pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta). \tag{7}$$

Check for $\theta' \neq \theta$.

Verify (7) for g, ψ and α in the theorem.

$$\pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta') = \pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta) \qquad (8)$$

Change variables in (8) from θ' to u at fixed θ using Eqn 1,

$$\pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta) = \pi(\theta)g(u)\alpha(\theta'(\theta,u)|\theta)dud\theta.$$

If $r(\theta',u'|\theta,u) = 1/r(\theta,u|\theta',u')$ then $r(\theta',u'|\theta,u) \leq 1$ WLOG.

$$\begin{aligned} RHS &= \pi(\theta)g(u)\frac{\pi(\theta')g(u')}{\pi(\theta)g(u)}J_\psi(\theta,u)dud\theta \\ &= \pi(\theta')g(u')\left|\frac{\partial(\theta',u')}{\partial(\theta,u)}\right|dud\theta \\ &= \pi(\theta')g(u')du'd\theta' \\ &= \pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta') \end{aligned}$$

making a change of variables and using $\alpha(\theta|\theta') = 1$.

We assumed

$$r(\theta', u'|\theta, u) = 1/r(\theta, u|\theta', u'), \tag{9}$$

so we could take $r(\theta', u'|\theta, u) \leq 1$ WLOG. Verify this. From (3)

$$1/r(\theta, u|\theta', u') = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta', u')^{-1}.$$

and

$$r(\theta', u'|\theta, u) = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u)$$

so we need $J_\psi(\theta', u')^{-1}\big|_{(\theta',u')=\psi(\theta,u)} = J_\psi(\theta, u)$. We know

$$J_\psi(\theta', u')^{-1}\big|_{(\theta',u')=\psi(\theta,u)} = J_{\psi^{-1}}(\theta, u).$$

Now $(\theta, u) = \psi(\psi(\theta, u))$ is useful: ψ is ψ^{-1}.

For measurable A, B in (6), the involution maps

$$\{(\theta, u) : \theta \in A,\ \psi_1(\theta, u) \in B\}$$

bijectively onto

$$\{(\theta', u') : \theta' \in B,\ \psi_1(\theta', u') \in A\},$$

so integration domains map correctly under the transforms. □

The Jacobian must be non-singular, so that the change of variables is well defined.

We certainly need $\dim(\theta', u') = \dim(\theta, u)$. This condition is called "dimension matching".

Example: "Random walk on a log scale". Suppose we are targeting $\theta \sim \exp(1)$ and we use the proposal

$$u \sim U(1/2, 2), \quad \theta' = u\theta \quad \text{so that} \quad (\theta', u') = (u\theta, 1/u)$$

$g(u) = 1/(2 - 0.5),\ 0.5 < u < 2$ and $\dim(\theta', u') = \dim(\theta, u) = 2$ so dimensions match. The Jacobian is $1/u$ since

$$\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right| = \begin{vmatrix} u & \theta \\ 0 & -1/u^2 \end{vmatrix} = 1/u.$$

The algorithm is as follows. If $X_t = \theta$ then

1) simulate $u \sim U(1/2, 2)$ and set $\theta' = u\theta$;

2) with probability

$$\alpha(\theta'|\theta) = \min\left\{1, \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)}\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right|\right\} = \min\left\{1, e^{-\theta'+\theta}u^{-1}\right\},$$

set $X_{t+1} = \theta'$ and otherwise $X_{t+1} = \theta$.

Factors of $g(u')/g(u)$ cancel in α. This proposal is useful if simulating a density which is peaked or diverges at a boundary. This whole framework is a useful restatement of MCMC.

Exercise: show that the Jacobian for the SRW proposal

$$u \sim U(-a, a), \qquad (\theta', u') = (\theta + u, -u)$$

is equal one.

Exercise: in the **swiss**-data regression example in lecture 10, I gave the MCMC updates for z and θ. The update for σ is

$$u \sim U(\delta, 1/\delta), \qquad \sigma' = u\sigma$$

with $0 < \delta < 1$ a constant we can adjust for efficient MCMC. Calculate the acceptance probability (answer in R-code).

Matched proposals

When we have a mixture of N kernels, detailed balance is

$$\sum_{i=1}^{N} \pi(d\theta)\xi_i K_i(\theta, d\theta') \;=\; \sum_{j=1}^{N} \pi(d\theta')\xi_j K_j(\theta', d\theta). \qquad \text{(A)}$$

with

$$K_i(\theta, d\theta') = q_i(d\theta'|\theta)\alpha_i(\theta'|\theta) + c_i(\theta)\delta_\theta(d\theta').$$

Up to now this held because

$$\pi(d\theta)\xi_i K_i(\theta, d\theta') \;=\; \pi(d\theta')\xi_i K_i(\theta', d\theta), \quad i = 1, \ldots, N \qquad \text{(B)}$$

as each kernel satisfied DB by itself. This isn't necessary.

Let $j = \sigma_i,\ i = \sigma_j$ pair kernels $K_i,\ K_j$. We get DB in (A) if

$$\pi(d\theta)\xi_i q_i(d\theta'|\theta)\alpha_i(\theta'|\theta) = \pi(d\theta')\xi_j q_j(d\theta|\theta')\alpha_j(\theta|\theta'). \qquad \text{(C)}$$

Here (C) takes care of DB in (A) for $\theta' \neq \theta$ and rejection terms cancel between $\xi_i K_i$ on each side of (A) as for (B).

For transforms, let $\sigma = (\sigma_1, \ldots, \sigma_N)$ map pairs and if $j = \sigma_i$,

$$\psi_i : \Omega \times U_i \longrightarrow \Omega \times U_j, \qquad (\theta', u') = \psi_i(\theta, u), \quad \psi_j = \psi_i^{-1}$$

are differentiable bijections.

Let $q_i(d\theta'|\theta) = g_i(u)du$ and $q_j(d\theta|\theta') = g_j(u')du'$. If

$$\begin{aligned} \alpha_i(\theta'|\theta) &= \min\left\{1, \ \frac{\pi(\theta')\,\xi_j\,q_j(\theta|\theta')}{\pi(\theta)\,\xi_i\,q_i(\theta'|\theta)}\right\} \\ &= \min\left\{1, \ \frac{\pi(\theta')\,\xi_{\sigma_i}\,g_{\sigma_i}(u')}{\pi(\theta)\,\xi_i\,g_i(u)}\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right|\right\} \end{aligned} \qquad \text{(D)}$$

then we satisfy (C). The proof is essentially the same as before.

Now *individual updates* need not be reversible as in (B).

We don't even need $q_i(\theta'|\theta) > 0 \Rightarrow q_i(\theta|\theta') > 0$. The chain as a whole is still reversible as (D) gives (C) and hence (A).

Eg: $\theta \sim \exp(1)$ with $u \sim U(1,2)$ wp 1/2 else $u \sim U(0.5,1)$

The mapping $(\theta', u') = (u\theta, 1/u)$ and Jacobian are unchanged.

The algorithm is

(1) with probability 1/2
    (a) set $u \sim U(1,2)$ and otherwise (b) set $u \sim U(0.5,1)$.

(2) Set $\theta' = u\theta$.

(3) if we chose (a) then

$$g_2(u')/g_1(u) = \frac{U(u'; 0.5, 1)}{U(u; 1, 2)} = 2.$$

Since $\rho_1 = \rho_2 = 1/2$, we accept θ' with probability

$$\alpha(\theta'|\theta) = \min\{1, 2 \times e^{-\theta'+\theta-\log(u)}\},$$

and if we chose (b) then $g_1(u')/g_2(u) = 0.5$ and accept θ' wp

$$\alpha(\theta'|\theta) = \min\{1, 0.5 \times e^{-\theta'+\theta-\log(u)}\}.$$